Array-based lists

part 2

Alina Shaikhet

Review

Implementations of the **List** interface

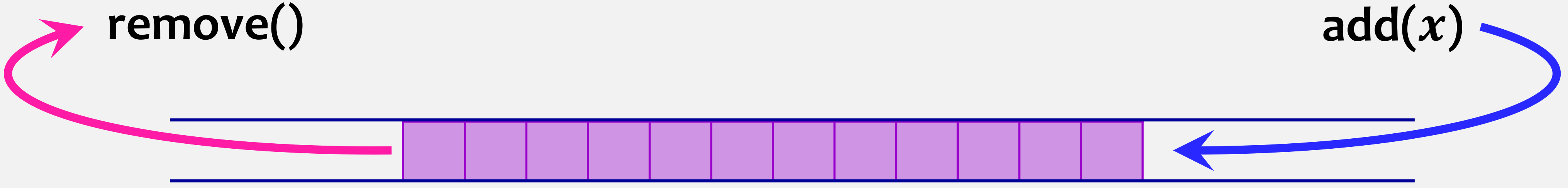
	get(i) / set(i,x)	add(i,x) / remove(i)
ArrayStack	$O(1)$	$O(1+n-i)$
LinkedList	$O(1+\min\{i,n-i\})$	$O(1+\min\{i,n-i\})$
ods ArrayDeque	$O(1)$	$O(1+\min\{i,n-i\})$

we have seen

today

FIFO Queue

FIFO Queue represents a sequence of FIFO elements.
We add to the end of the queue and remove from the front.



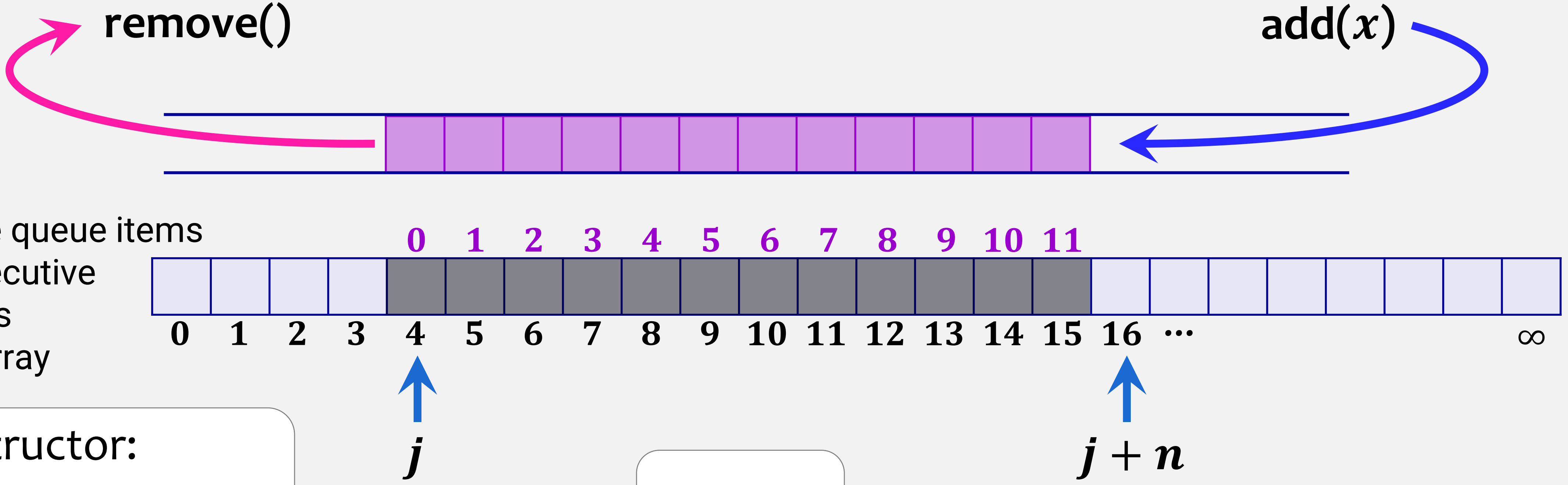
size()
add(x)
remove() – remove and return the "oldest" element

We can use **ArrayList** implementation and add/remove only to front/back. However, this won't get us our desired $O(1)$ time for add/remove operations.

add(x)	**ArrayList.add(size(), x)**	or	**ArrayList.add(0, x)**
remove()	**ArrayList.remove(0)**		**ArrayList.remove(size() − 1)**

To avoid shifting we will use "pointer" to the front index of our queue elements (j)

FIFO Queue



we store queue items at consecutive locations in this array

constructor:

$$a = \textbf{new T}[\infty];$$
$$n = 0;$$
$$j = 0;$$

T[] a;
int n;
int j;

index at which queue begins

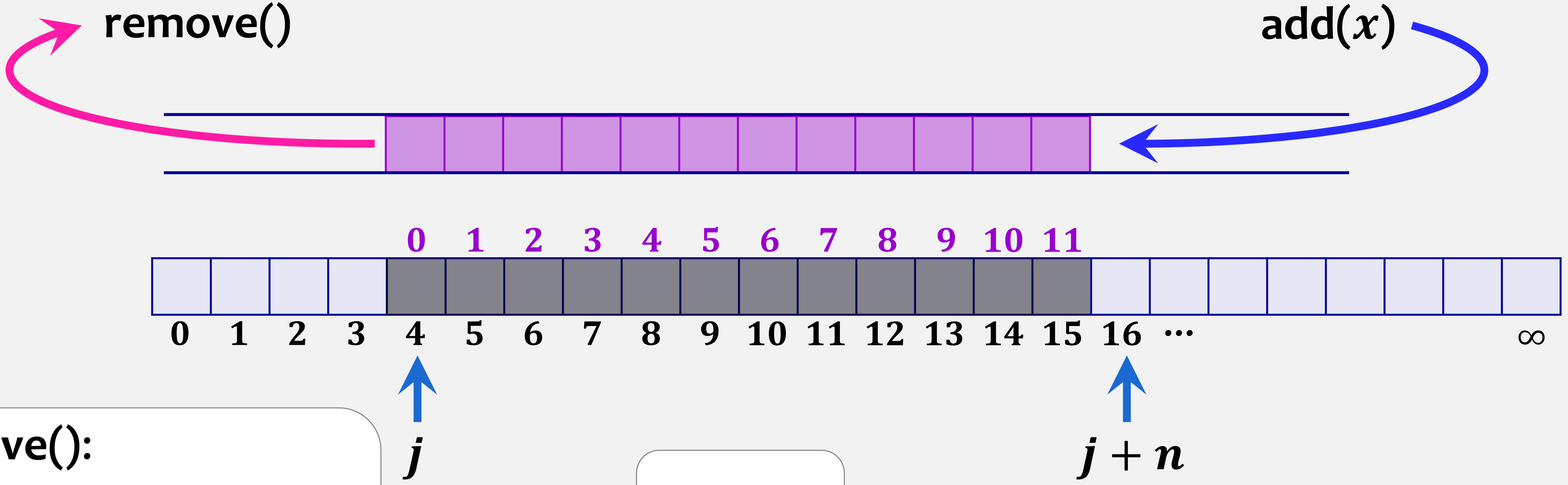
add(x) – which position do we add?

it is no longer at position n

in our example $n = 12,\ j = 4$

Alina Shaikhet – COMP2402 – Carleton University

FIFO Queue



remove():

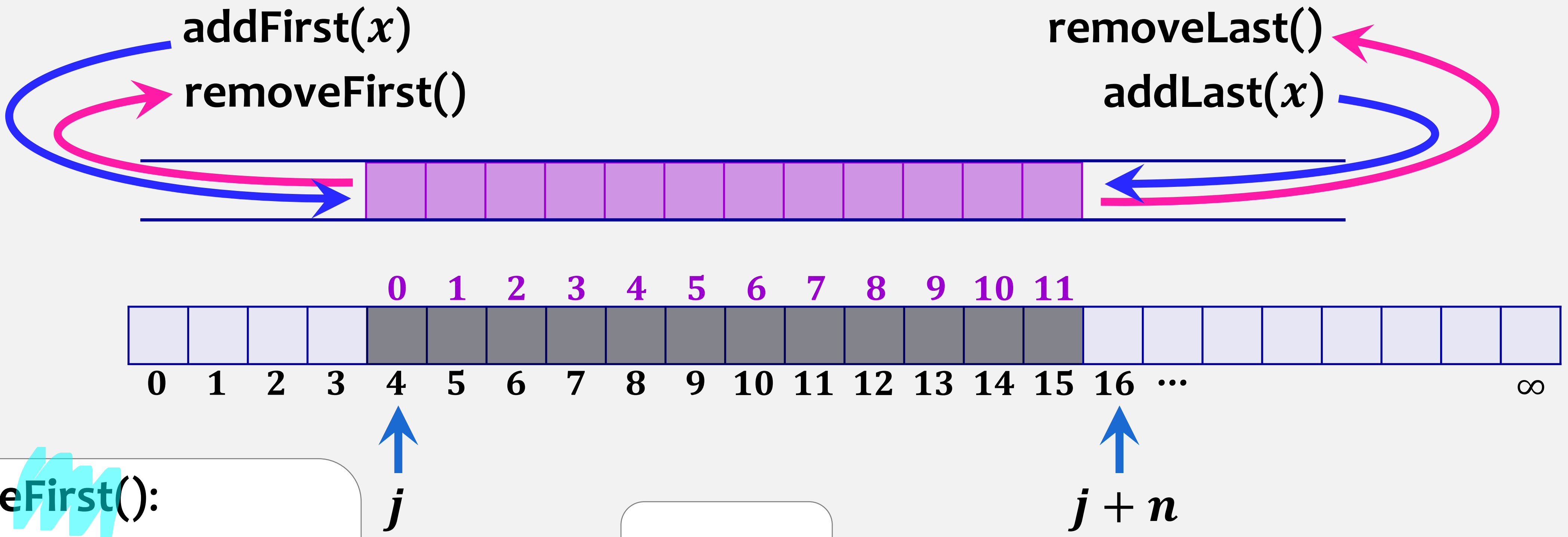
if $(n == 0)$ throw exception
T $x = a[j]$;
$a[j] = \mathbf{null}$;
$j++$;
$n--$;
return x;

T[] a;
int n;
int j;

add(x):

$a[j+n] = x$;
$n++$;

Deque



removeFirst():

if $(n == 0)$ throw exception
T $x = a[j]$;
$a[j] = \mathbf{null}$;
$j++$;
$n--$;
return x;

T[] a;
int n;
int j;

addLast(x):

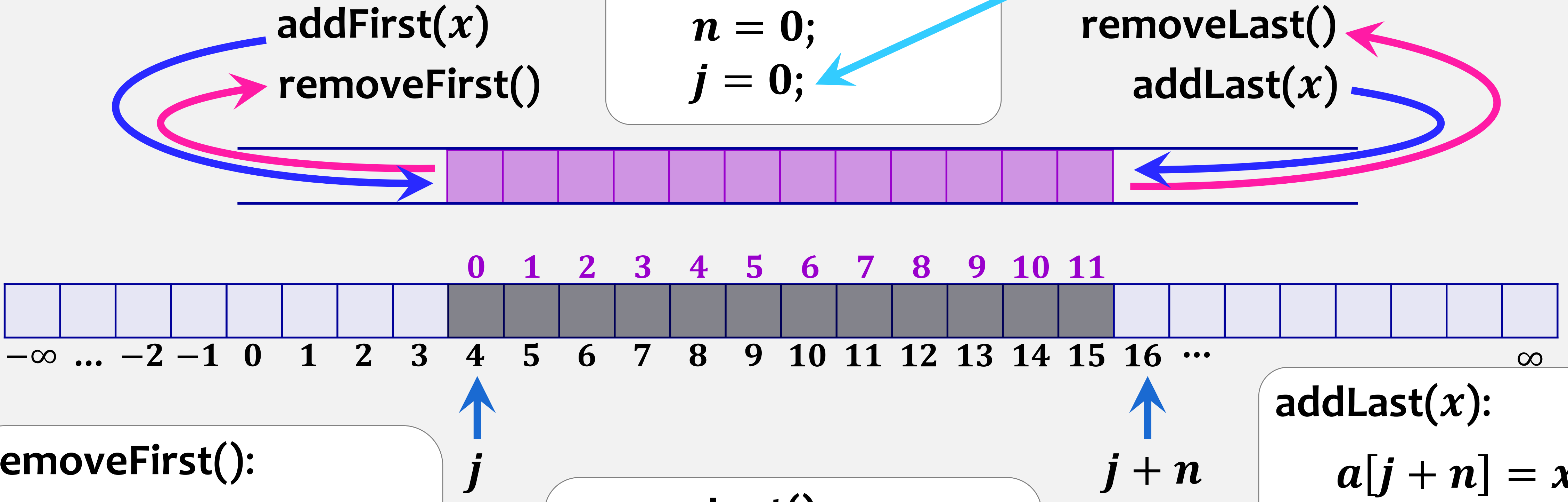
$a[j+n] = x$;
$n++$;

Deque

constructor:

$a = \text{new T}[\infty];$
$n = 0;$
$j = 0;$

but you can choose any other index



addLast(x):

$a[j+n] = x;$
$n++;$

removeFirst():

if $(n == 0)$ throw exception
T $x = a[j];$
$a[j] = \text{null};$
$j++;$
$n--;$
return x;

removeLast():

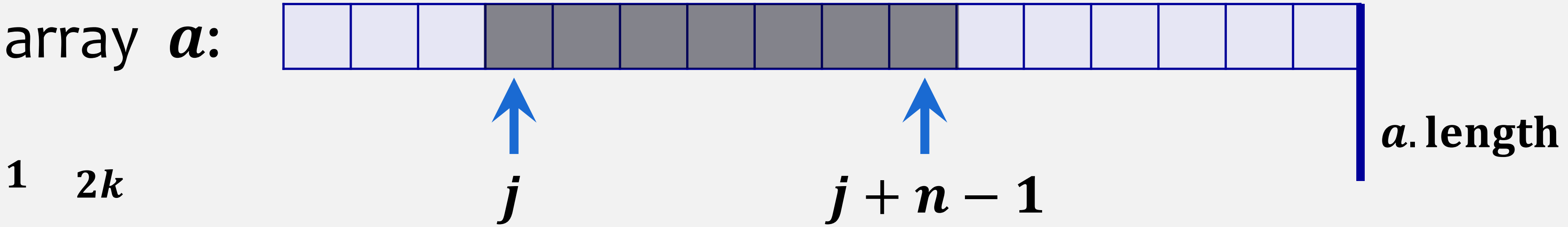
if $(n == 0)$ throw exception
T $x = a[j+n-1];$
$a[j+n-1] = \text{null};$
$n--;$
return x;

addFirst(x):

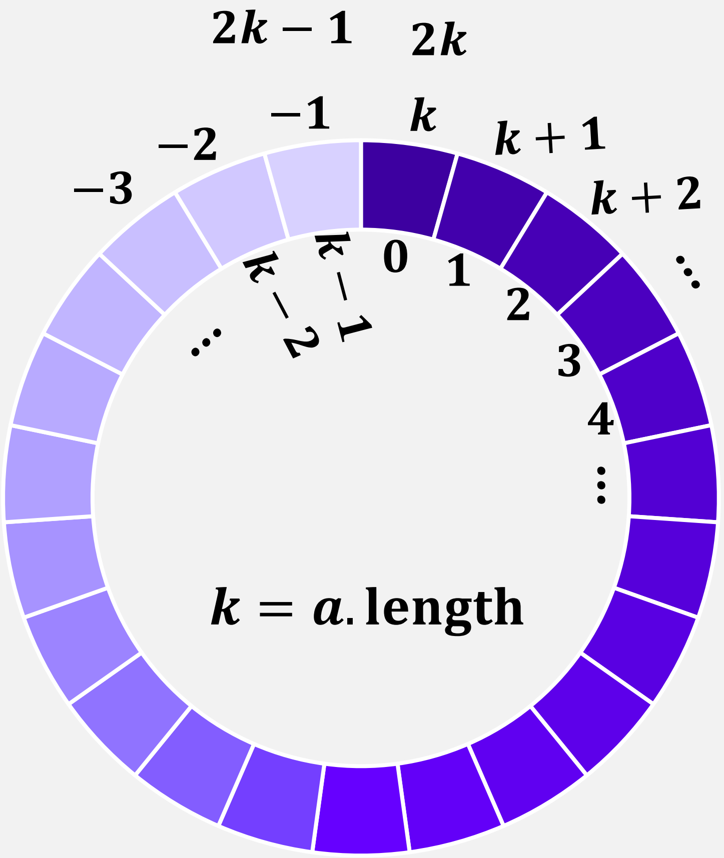
$j--;$
$a[j] = x;$
$n++;$

Circular Array & MOD

In Java we write $i\%k$. Modulus operator (%) returns the division remainder.



j – the first index
n – number of elements in the queue



circular array

$i \bmod k \in \{0, \dots, k-1\}$

$i \bmod k = r$ remainder

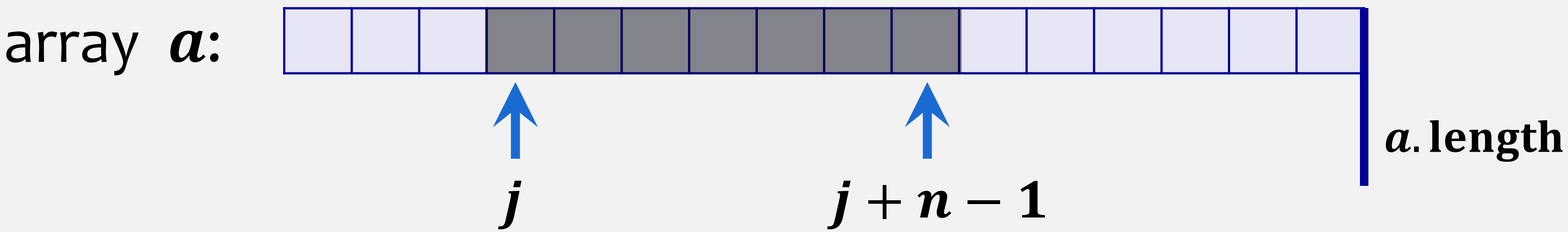
$r \in \{0, \dots, k-1\}$

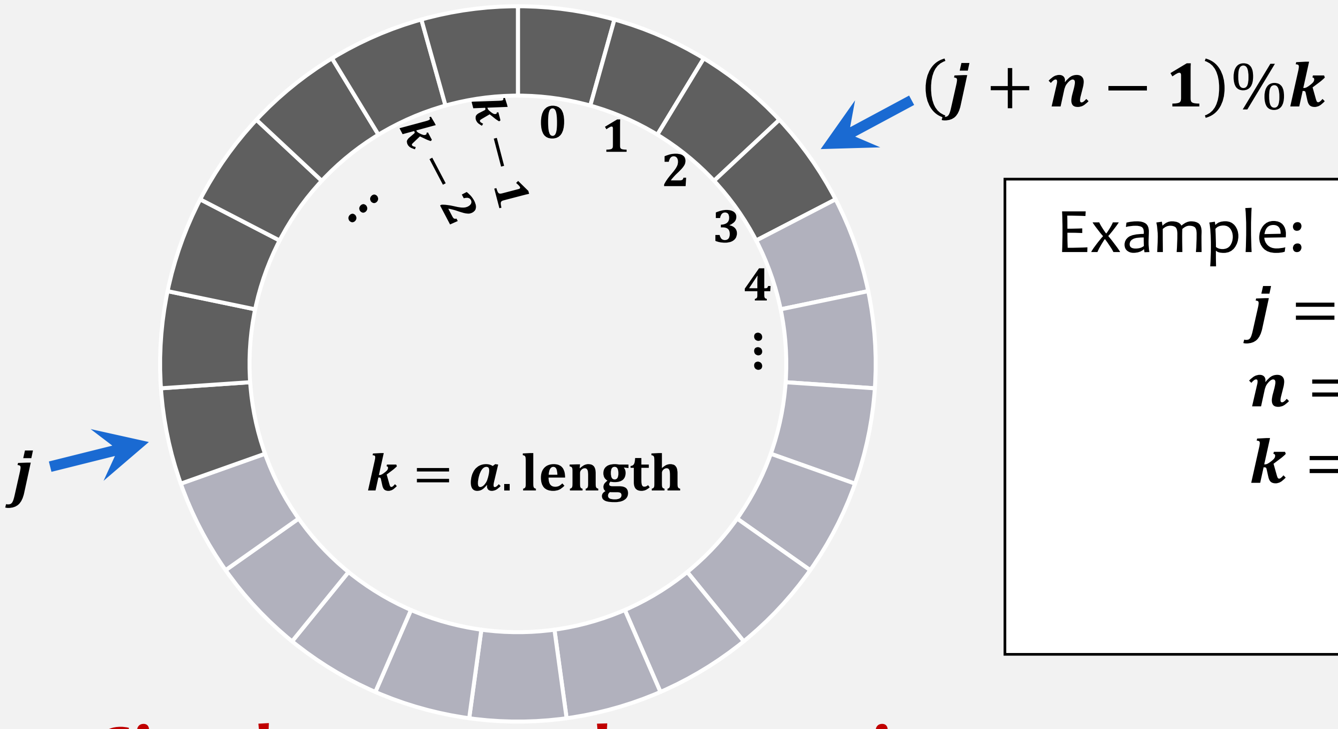
some integer

$i = p \cdot k + r$

$i \bmod k = i - p \cdot k$

Circular Array & MOD





Example:

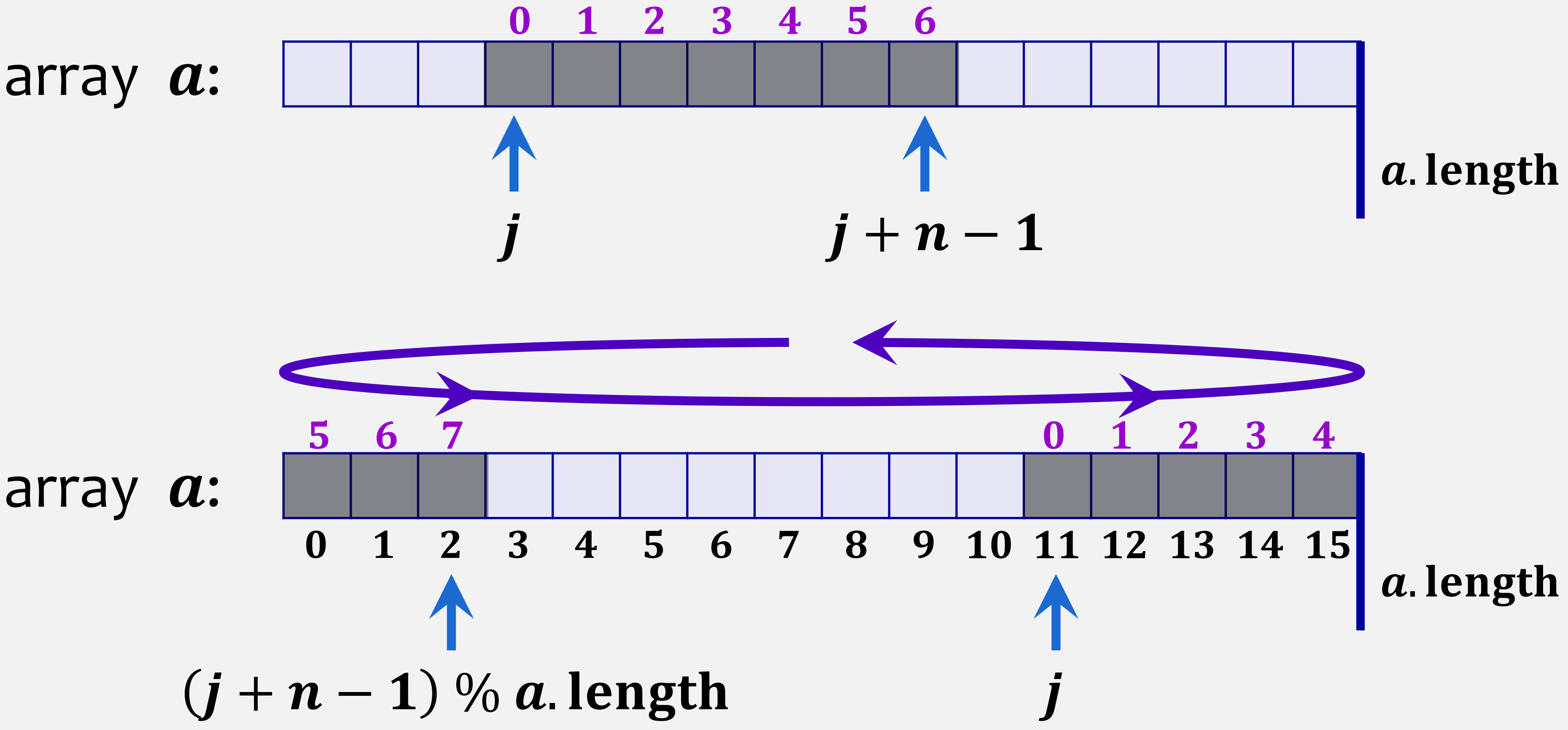
$$j = 17 \qquad n = 11 \qquad k = 24$$

$$(j+n-1)\%k$$
$$(17+11-1)\%24$$
$$27\%24$$
$$3$$

$$i \bmod k = i - p \cdot k$$

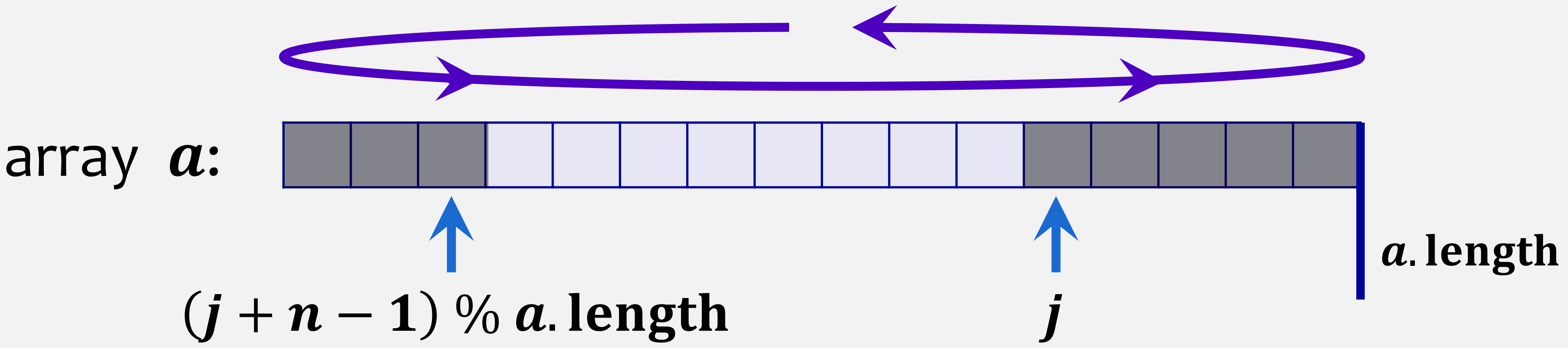
Circular arrays do not exist

Circular Array & MOD



ArrayDeque in JCF

Deque



constructor:

$a = \text{new } \mathrm{T}[1];$
$n = 0;$
$j = 0;$

We need to guarantee that j is a valid index

addFirst(x):

if $(n+1 > a.\mathrm{length})$ then **resize()**;
$j = (j == 0)\ ?\ a.\mathrm{length} - 1 : j - 1;$
$a[j] = x;$
$n++;$

ignoring **resize()** $O(1)$

removeFirst():

if $(n == 0)$ throw exception
T $x = a[j];$
$a[j] = \mathrm{null};$
$j = (j+1)\ \%\ a.\mathrm{length};$
$n--;$
if $(3n \le a.\mathrm{length})$: **resize()**;
return x;

removeLast():

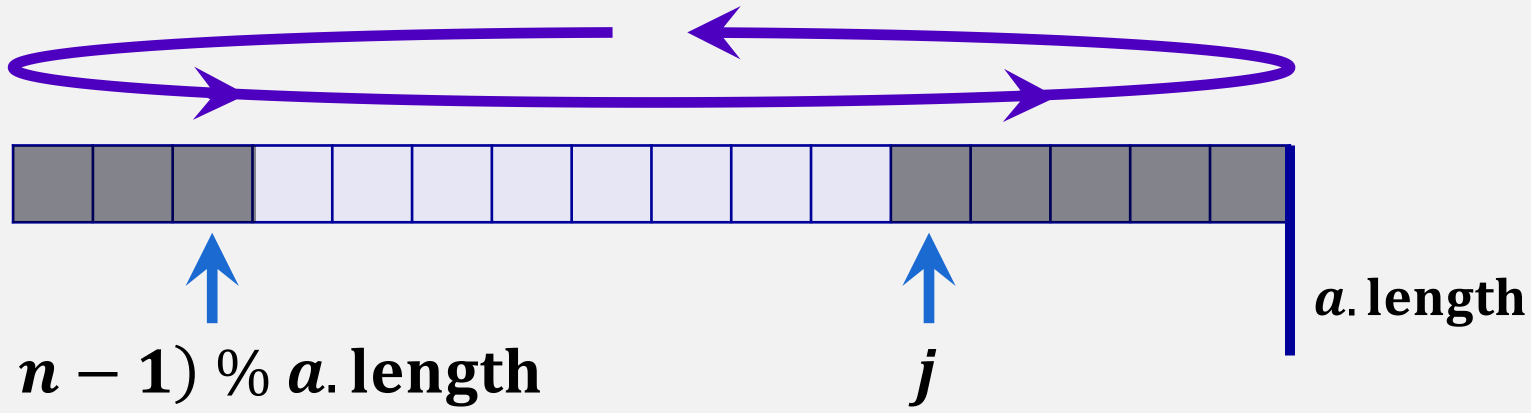
if $(n == 0)$ throw exception
T $x = a[(j+n-1)\%\ a.\mathrm{length}];$
$a[(j+n-1)\%\ a.\mathrm{length}] = \mathrm{null};$
$n--;$
if $(3n \le a.\mathrm{length})$: **resize()**;
return x;

addLast(x):

if $(n+1 > a.\mathrm{length})$ then **resize()**;
$a[(j+n)\%\ a.\mathrm{length}] = x;$
$n++;$

Alina Shaikhet – COMP2402 – Carleton University

FIFO Queue



constructor:

$a = \text{new T}[1];$
$n = 0;$
$j = 0;$

Refer to ods textbook:
ArrayQueue implements the FIFO Queue interface

remove():

if $(n == 0)$ throw exception
T $x = a[j];$
$a[j] = \text{null};$
$j = (j + 1)\ \%\ a.\text{length};$
$n - -;$
if $(3n \le a.\text{length})$: **resize();**
return x;

ignoring **resize()** $O(1)$

add(x):

if $(n + 1 > a.\text{length})$ then **resize();**
$a[(j + n)\%\, a.\text{length}] = x;$
$n + +;$

resize()

to avoid 0-length array

void **resize():**

$$T[]\ \boldsymbol{b} = \text{new array}(\mathbf{max}\{2\boldsymbol{n}, \mathbf{1}\})$$

executes $\boldsymbol{n}$ times

$$\text{for } (\boldsymbol{j} = \mathbf{0}; \boldsymbol{i} < \boldsymbol{n};\ \boldsymbol{i}++)$$

$$\boldsymbol{b}[\boldsymbol{i}] = \boldsymbol{a}[(\boldsymbol{j} + \boldsymbol{i})\%\boldsymbol{a}.\mathbf{length}];$$

$$\boldsymbol{a} = \boldsymbol{b};$$

$$\boldsymbol{j} = \mathbf{0};$$

$$\boldsymbol{O}(\boldsymbol{n})$$

Theorem 2.2

An **ArrayQueue** implements the (FIFO) Queue interface. Ignoring the cost of calls to **resize()**, an **ArrayQueue** supports the operations **add**($\boldsymbol{x}$) and **remove()** in $\boldsymbol{O}(\mathbf{1})$ time per operation. Furthermore, beginning with an empty **ArrayQueue**, any sequence of $\boldsymbol{m}$ **add**($\boldsymbol{x}$) and **remove()** operations results in a total of $\boldsymbol{O}(\boldsymbol{m})$ time spent during all calls to **resize()**.

Review

Implementations of the **List** interface

	get(i) / set(i,x)	add(i,x) / remove(i)
ArrayStack	$\boldsymbol{O(1)}$	$\boldsymbol{O(1+n-i)}$
LinkedList	$\boldsymbol{O(1+\min\{i,n-i\})}$	$\boldsymbol{O(1+\min\{i,n-i\})}$
ods ArrayDeque	$\boldsymbol{O(1)}$	$\boldsymbol{O(1+\min\{i,n-i\})}$

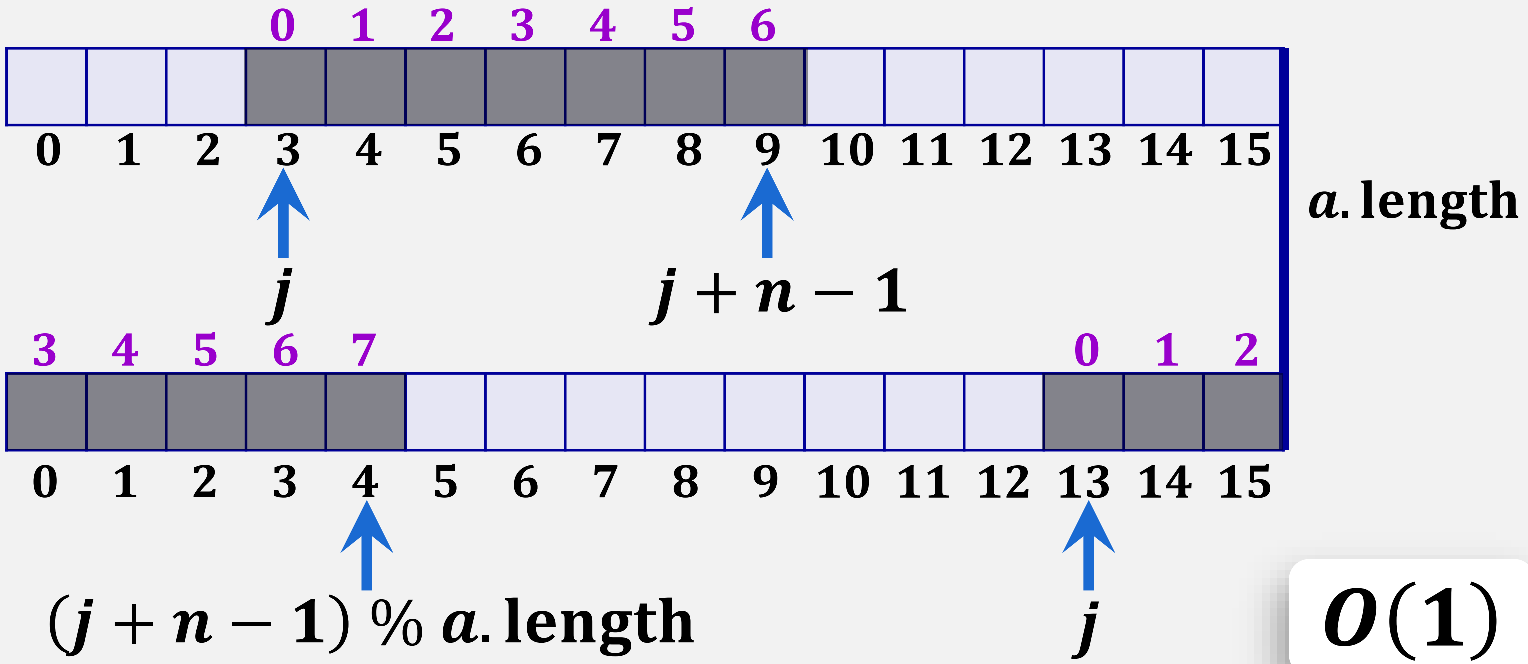
we have seen

today

Alina Shaikhet – COMP2402 – Carleton University

ArrayDeque

ArrayDeque implements the **List** interface using circular array with $O(1)$ amortized **Deque** operations (adding and removing to both ends of our sequence).



$O(1)$

T[] a;
int n;
int j;

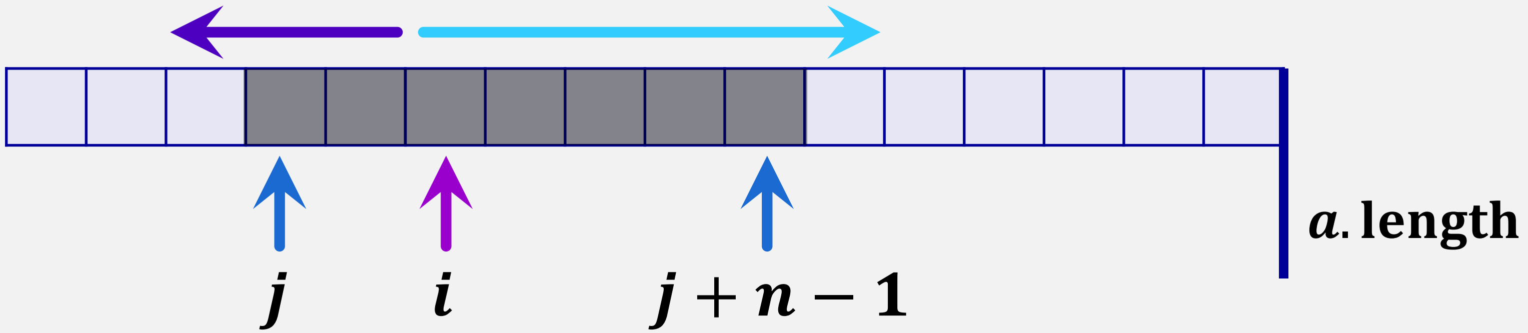
constructor:

$a = \mathbf{new\ T}[1];$
$n = 0;$
$j = 0;$

get(i): check bounds;
return $a[(j + i)\ \%\ a.\mathbf{length}];$

set(i,x): check bounds;
T $y = a[(j + i)\ \%\ a.\mathbf{length}];$ //T $y = \mathrm{get}(i);$
$a[(j + i)\ \%\ a.\mathbf{length}] = x;$
return y;

ArrayDeque



add(i,x): if $(n+1 > a.\mathbf{length})$ then **resize()**;

if $\left(i < \frac{n}{2}\right)$ then // i is in the first half

shift to the left — $O(i)$

$j = (j == 0)\ ?\ a.\mathbf{length} - 1 : j - 1;$ // move j left   // $j = (j-1)$ % a

for $(k = 0;\ k \le i - 1;\ k++)$

$a[(j+k)\ \%\ a.\mathbf{length}] = a[(j+k+1)\ \%\ a.\mathbf{length}];$

else // i is in the second half

shift to the right — $O(n-i)$

for $(k = n;\ k > i;\ k--)$

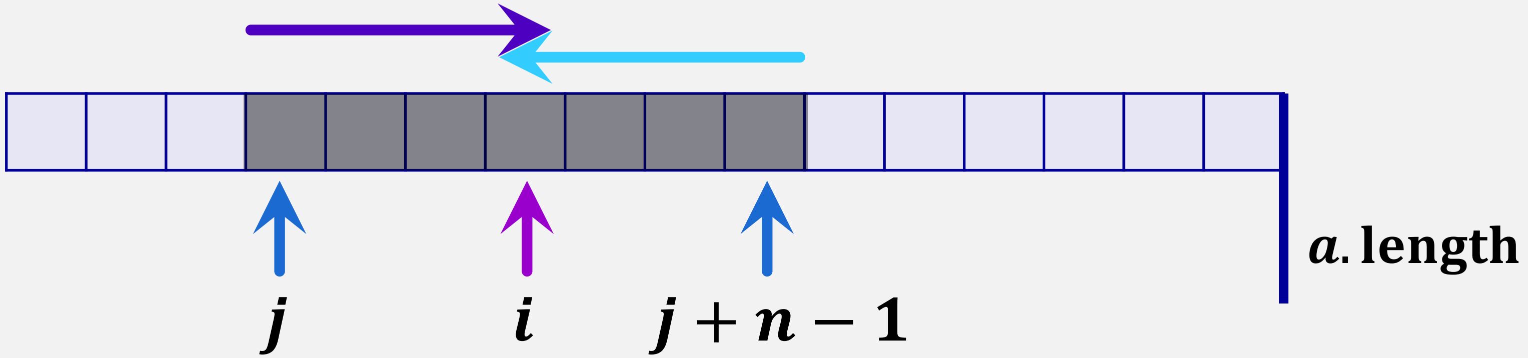
$a[(j+k)\ \%\ a.\mathbf{length}] = a[(j+k-1)\ \%\ a.\mathbf{length}];$

$a[(j+i)\ \%\ a.\mathbf{length}] = x;$

$n++;$

Running time: $\boxed{O(1+\min(i, n-i))}$ **amortized**

ArrayDeque



remove(i,x): $x = a[(j+i)\ \%\ a.\mathbf{length}]$;

if $\left(i < \frac{n}{2}\right)$ then // shift $a[0],\ldots,[i-1]$ right one position

$\boxed{O(i)}$ shift $\begin{cases} \text{for } (k = i;\ k > 0;\ k--) \\ \quad a[(j+k)\ \%\ a.\mathbf{length}] = a[(j+k-1)\ \%\ a.\mathbf{length}]; \end{cases}$

$j = (j+1)\ \%\ a.\mathbf{length}$ // move j right

else // shift $a[i+1],\ldots,a[n-1]$ left one position

$\boxed{O(n-i)}$ shift $\begin{cases} \text{for } (k = i;\ k < n-1;\ k++) \\ \quad a[(j+k)\ \%\ a.\mathbf{length}] = a[(j+k+1)\ \%\ a.\mathbf{length}]; \end{cases}$

$n--$;

if $(3n < a.\mathbf{length})$ then **resize()**;

return x;

Running time: $\boxed{O\big(1+\min(i, n-i)\big)}$ **amortized**

Theorem 2.3

An **ArrayDeque** implements the **List** interface. Ignoring the cost of calls to **resize()**, an **ArrayDeque** supports the operations

- **get**$(\boldsymbol{i})$ and **set**$(\boldsymbol{i}, \boldsymbol{x})$ in $\boldsymbol{O}(\mathbf{1})$ time per operation; and
- **add**$(\boldsymbol{i}, \boldsymbol{x})$ and **remove**$(\boldsymbol{i})$ in $\boldsymbol{O}(\mathbf{1} + \mathbf{min}\{\boldsymbol{i}, \boldsymbol{n} - \boldsymbol{i}\})$ time per operation.

Furthermore, beginning with an empty **ArrayDeque**, performing any sequence of $\boldsymbol{m}$ **add**$(\boldsymbol{i}, \boldsymbol{x})$ and **remove**$(\boldsymbol{i})$ operations results in a total of $\boldsymbol{O}(\boldsymbol{m})$ time spent during all calls to **resize()**.